

Giovanni Sambo

MSc Aeronautical Engineering — Aerodynamics & CFD

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Summary

MSc Aeronautical Engineering candidate at Politecnico di Milano (graduation Oct 2026), specializing in experimental aerodynamics, CFD, and wind energy systems. Hands-on wind tunnel testing experience, aeroelastic design, and industrial flight testing. Available Oct 2026, willing to relocate across Europe.

Education

- 2023 - 2026 **MSc Aeronautical Engineering**, *Politecnico di Milano*, Milan, Italy
Aerodynamics, CFD, experimental fluid dynamics, wind & energy systems, flight testing, propulsion.
- Jan - Jun 2024 **Erasmus Exchange**, *ETSi, Univ. de Sevilla*, Seville, Spain
Heat Transfer, Aircraft Trajectory Optimization.
- 2020 - 2023 **BSc Aerospace Engineering**, *Politecnico di Milano*, Milan, Italy

Master's Thesis (ongoing)

Title *Experimental and Numerical Study of the Flettner Rotor*

Wind tunnel campaign (PoliMi) on a rotating cylinder exploiting the Magnus effect for wind-assisted ship propulsion (WASP). Four endplate configurations, varying Re and tip-speed ratios. Force measurements (C_L , C_D), smoke flow visualization, CFD validation.

Academic Projects

- 2024 - 2025 **Flight Testing — Pipistrel Virus SW121A**, *Politecnico di Milano & Pipistrel Italia*
Industrial campaign: flight card design and mission planning, DEWESoft DAS, in-flight data acquisition, post-flight analysis of stall and performance.
- 2024 - 2025 **2.3 MW IEC Class IA Wind Turbine Design**, *Politecnico di Milano*
Aeroelastic design in C_p - Λ (BEM). Structural sizing via BECAS/RotorCS, TurbSim wind fields (IEC 61400-1), PID gain-scheduling, DLC load assessment. CoE minimization over 8 design variants.
- 2024 - 2025 **CFD Analysis — Formula 1 Front Wing**, *Politecnico di Milano*
2D compressible RANS/URANS in SU2. Ground effect and porpoising (6 Hz). SA vs SST $k-\omega$. Post-processing in Paraview.
- 2024 - 2025 **Experimental Fluid Dynamics — Wind Tunnel Labs**, *Politecnico di Milano*
Three **wind tunnel** campaigns (PoliMi): force balance on NACA 0015; pressure and wake survey on NACA 23012; PIV of side-by-side eVTOL rotor configurations.
- 2024 - 2025 **Aerodynamic Analysis of a Travel Bus**, *Politecnico di Milano*
Drag reduction via OpenFOAM, validated against **wind tunnel** measurements (PoliMi). Combined CFD and experimental approach.
- 2024 - 2025 **Crosswind Rollover — Heavy Truck**, *Politecnico di Milano*
Multi-DOF MATLAB model with Pacejka tyres under transient aerodynamic crosswind loading.

Work Experience

- Jun - Jul 2019 **Assistant Aircraft Designer — Internship**, *Superjet Sukhoi*
Maintenance documentation and data archiving for the Superjet Sukhoi 100.
- 2018 - 2022 **Other roles**, *Turismo Srl · Air Food Srl · Gattopardo*
Lifeguard, salesperson, waiter. Communication and reliability in fast-paced environments.

Technical Skills

CFD & Aero	SU2, OpenFOAM, XFOIL, Paraview, Gmsh	Wind Turbine	C_p - Λ , BECAS, RotorCS, TurbSim
Experimental	Wind tunnel, PIV, force balance, DEWE-Soft	Programming	MATLAB (advanced), C (basic), Excel
CAD	SolidWorks, Inventor, Solid Edge		

Languages

Italian (native) English (professional) Spanish (basic)